Implementation and Evaluation Toward Distributed Inference of Large-Scale SNNs Using BindsNET

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1. Summary of the Research

Spiking neural networks (SNNs) process information using spikes. This study uses the Diehl and Cook model implemented with BindsNET as the baseline. The goal is to partition a large-scale SNN across multiple Raspberry Pi devices for distributed inference. Single-node and distributed inference will be compared in terms of time, communication, correctness, and scalability.

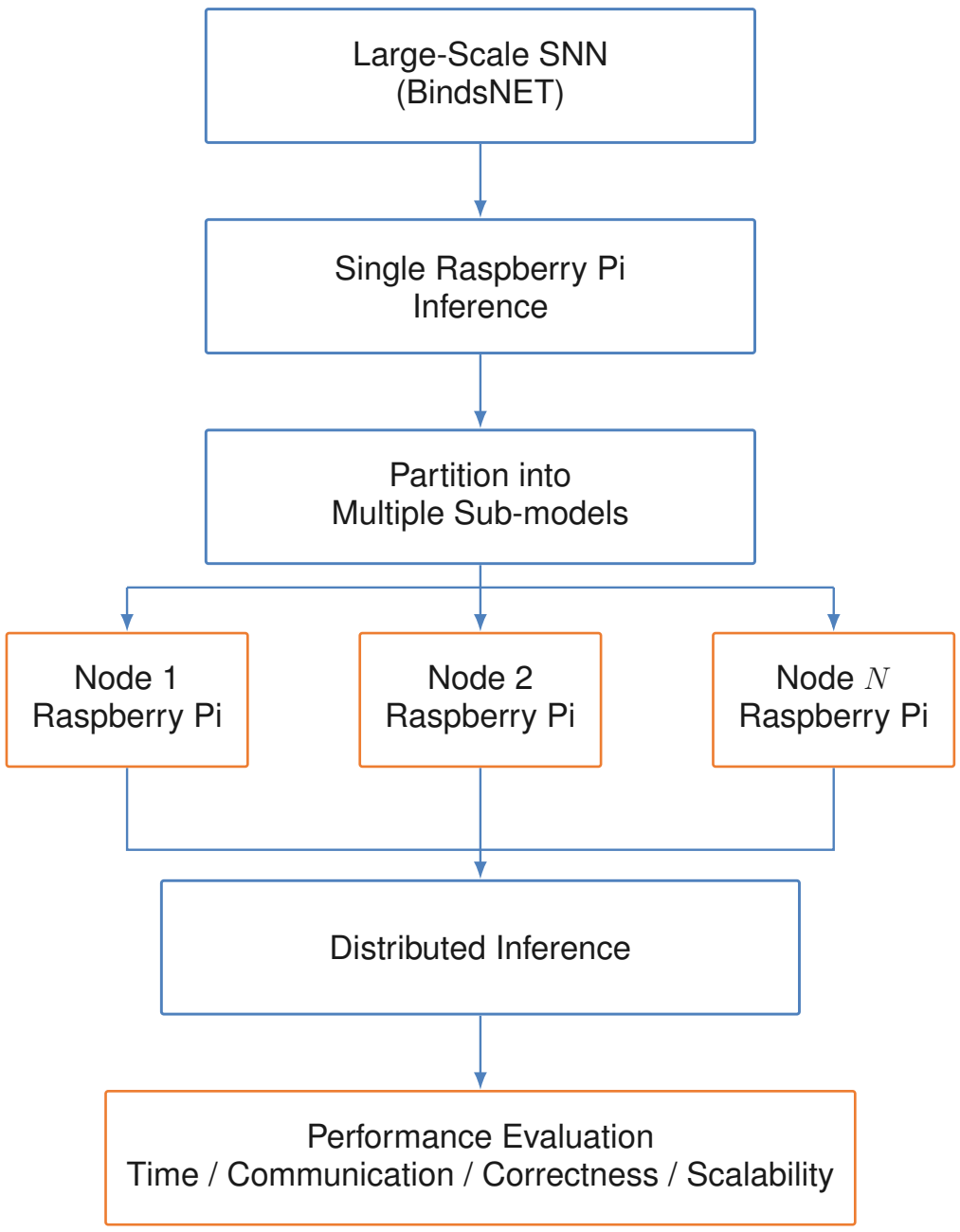

Fig. 1. Research concept for distributed SNN inference.

Evaluation Plan

Baseline	One Raspberry Pi
Distributed	Multiple Raspberry Pis
Input / Model	Same input condition and model scale
Metrics	Time, communication, correctness, scalability

2. SNN Model

The SNN is based on the Diehl and Cook model [1] and implemented using BindsNET, a Python library for building and simulating SNNs [2]. MNIST images are converted into spike trains, and input synaptic weights are learned using spike-timing-dependent plasticity (STDP). Inhibitory neurons regulate network activity and promote competition among neurons.

3. Current Progress

A 1,000-neuron SNN was first implemented and tested on CPU. The model was then scaled up to 10,000 neurons. A preliminary 10-part computation test matched the original computation under the tested conditions. The next step is to run the model on a Raspberry Pi and proceed toward distributed inference.

Stage	Status
1,000-neuron SNN	Completed
10,000-neuron SNN	Completed
Preliminary 10-part test	Matched
Single Raspberry Pi	Next step
Distributed inference	Future work

4. Remaining Tasks and Schedule

October: Run the large-scale SNN on one Raspberry Pi and confirm baseline inference.

November: Partition the model and deploy it to multiple Raspberry Pi devices.

December: Run distributed inference and measure time and communication overhead.

January: Increase the number of nodes and evaluate scalability.

Final stage: Compare single-node and distributed inference and summarize the results.

References

[1] P. U. Diehl and M. Cook, “Unsupervised learning of digit recognition using spike-timing-dependent plasticity,” *Frontiers in Computational Neuroscience*, vol. 9, Art. 99, 2015.
[2] H. Hazan et al., “BindsNET: A machine learning-oriented spiking neural networks library in Python,” *Frontiers in Neuroinformatics*, vol. 12, Art. 89, 2018.